

# Exploratory Data Analysis: Player Similarity Features

## Overview

This report identifies discriminative features for player similarity from two complementary datasets:

- **SkillCorner**: Tracking data (10Hz positions) from 10 A-League matches, 237 players
- **StatsBomb**: Event data from 3,464 matches across multiple competitions

The analysis covers movement intensity, spatial positioning, action profiles, and team interaction patterns.

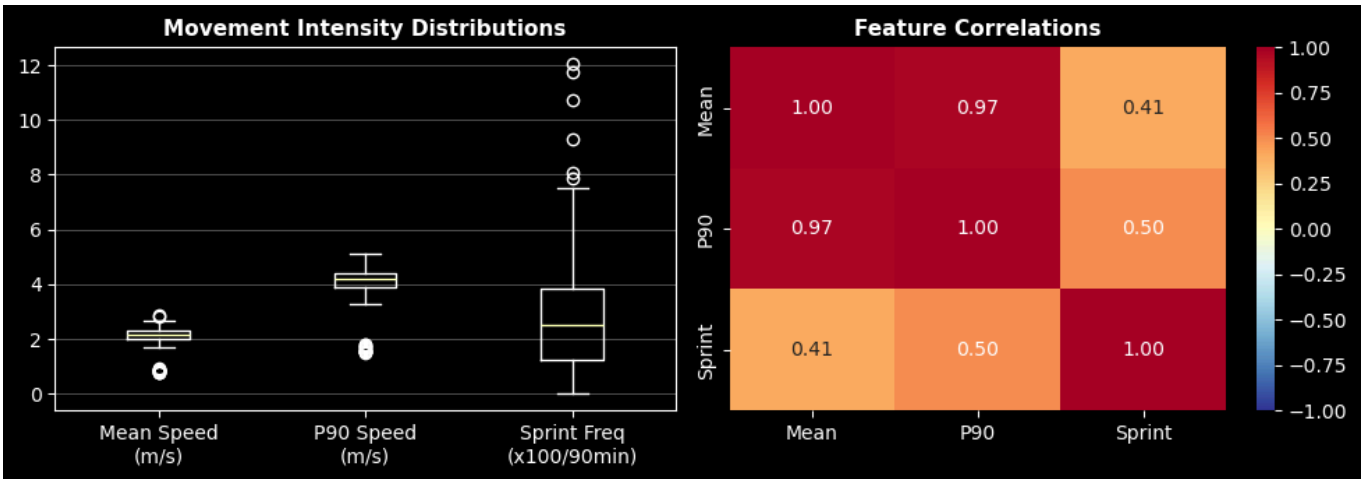
SkillCorner: 10 matches  
StatsBomb: 3464 matches

## Section 1: Movement Intensity (SkillCorner)

**Objective:** Assess whether speed-based metrics differentiate players and identify feature redundancy.

**Features:** Mean speed, 90th percentile speed (P90), and sprint frequency (frames > 7 m/s per 90 minutes).

Analyzed 206 players



Key correlations: Mean-P90: 0.973, Mean-Sprint: 0.405

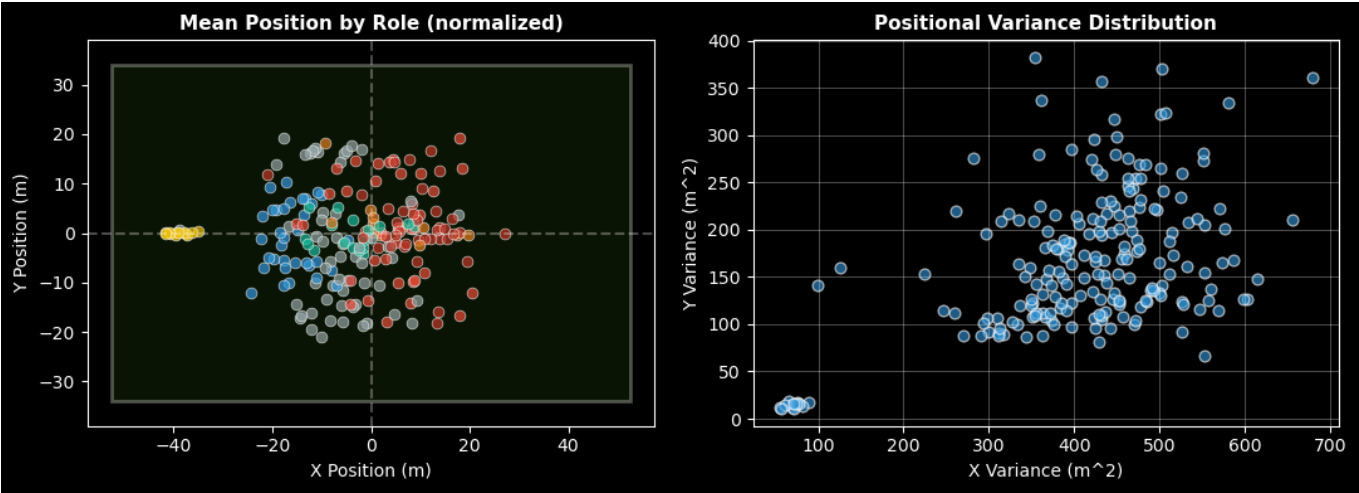
**Findings:** 206 players show substantial variation across all metrics. Mean and P90 speed are highly correlated ( $r \sim 0.97$ ), but sprint frequency captures distinct explosive behavior ( $r \sim 0.40$ - $0.50$  with speed metrics). All three features provide complementary signal for similarity.

## Section 2: Spatial Footprint (SkillCorner)

**Objective:** Analyze positional behavior to distinguish tactical roles.

**Features:** Mean position (x, y), positional variance, zone occupancy. Team-aware normalization ensures all players appear attacking toward +x.

Analyzed 206 players



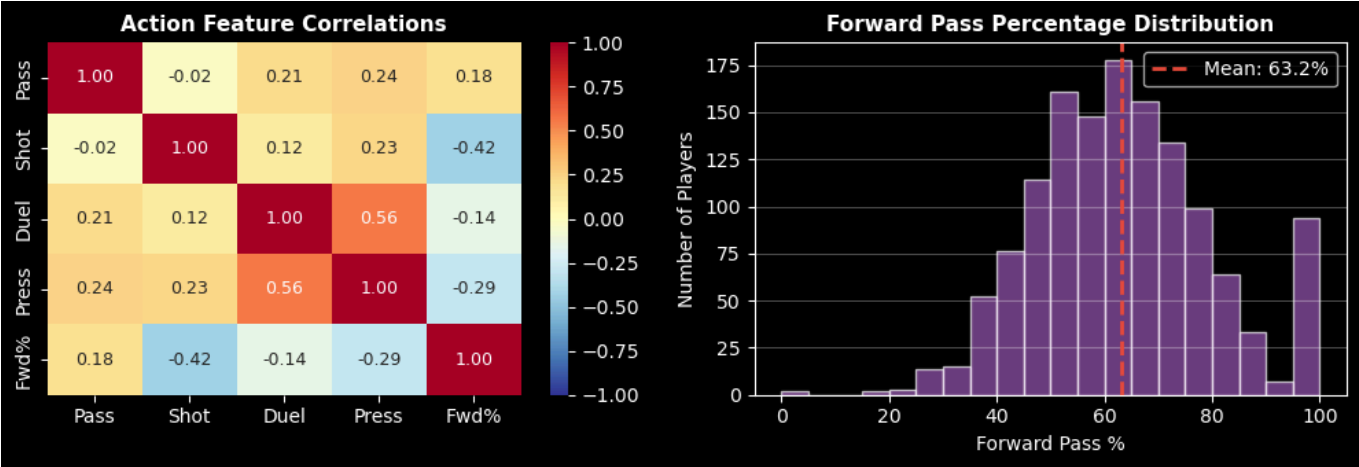
**Findings:** After team-aware normalization, clear spatial clustering emerges by tactical role. Goalkeepers occupy negative x, forwards positive x. Variance ranges from 500-1600 m<sup>2</sup> (x) and 10-300 m<sup>2</sup> (y), distinguishing roaming midfielders from positionally disciplined defenders.

### Section 3: Action Profiles (StatsBomb)

**Objective:** Characterize on-ball behavior through event frequencies.

**Features:** Actions per 90 (pass, shot, duel, pressure), forward pass percentage, action diversity (entropy).

Analyzed 1352 players from 200 matches



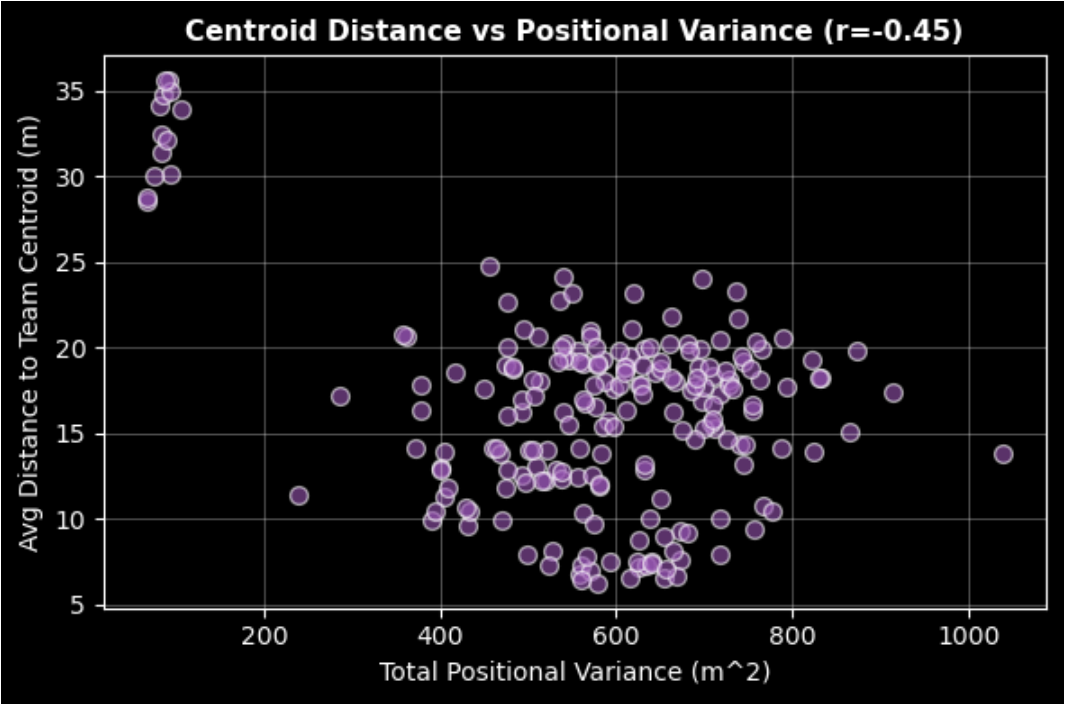
**Findings:** 1,352 players analyzed. Action features show low inter-correlations, indicating each captures distinct behavioral aspects. Forward pass percentage ranges from ~50-90%, distinguishing progressive playmakers from possession recyclers.

### Section 4: Interaction Signal (SkillCorner)

**Objective:** Capture relational positioning through team-relative metrics.

**Feature:** Average distance to team centroid - measures how far players position from their team's center of mass.

Analyzed 206 players



**Findings:** Moderate positive correlation between centroid distance and positional variance suggests these capture related but not identical information. Wide players and forwards show larger centroid distances. This feature provides complementary relational context.

### Summary: Feature Set for Player Similarity

| Category    | Features                                | Discriminative Power               |
|-------------|-----------------------------------------|------------------------------------|
| Movement    | Mean speed, P90 speed, Sprint frequency | High (distinct explosive patterns) |
| Spatial     | Mean position, Variance, Zone occupancy | High (role separation)             |
| Action      | Actions/90, Forward pass %, Diversity   | High (behavioral signatures)       |
| Interaction | Centroid distance                       | Moderate (relational context)      |

**Key Insights:**

- 1. **Non-redundancy:** Mean/P90 speed correlate ( $r=0.97$ ), but sprint frequency captures distinct patterns ( $r\sim0.4-0.5$ ). Action features show low inter-correlations.
- 2. **Role separation:** Normalized spatial features cleanly distinguish tactical positions.
- 3. **Multi-modal value:** Tracking (SkillCorner) and event (StatsBomb) data capture complementary aspects - movement/positioning vs. on-ball actions.

**Recommendation:** Combine all feature categories for comprehensive player similarity, with appropriate normalization for scale differences.